Arash Jalil Khabbazi, MS

PhD Student, Mechanical Engineering, Purdue University

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Research interests

I work on machine learning, optimization, and advanced control strategies for intelligent buildings and distributed energy resources.

Education

Purdue University, IN, United States — 2023 – Present
PhD in Mechanical Engineering — **GPA:** 4.0/4.0
(Minor: Computational Science and Engineering)
Adviser: Kevin J. Kircher

University of British Columbia, BC, Canada — 2021 – 2023
MS in Mechanical Engineering — **GPA:** 4.33/4.33 (94%)
Thesis: *Mixing gaseous hydrogen into natural gas distribution pipelines*
Adviser: Sunny Ri Li

University of Tabriz, EA, Iran — 2016 – 2020
BS in Mechanical Engineering — **GPA:** 4.0/4.0 (19.12/20) — **Rank:** 1/155+
Thesis: *Thermodynamic analysis of Kalina cycle system 11*
Adviser: S. Mohammad S. Mahmoudi

Publications

Journal articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, "Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review," *Applied Energy*, 2025. (doi)

1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, "Mixing hydrogen into natural gas distribution pipeline system through Tee junctions," *International Journal of Hydrogen Energy*, 2024. (doi)

Conference proceedings

4. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, K.J. Kircher, "What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?," *International High Performance Buildings Conference*, 2024, pp. 1–10. (doi)

3. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, "Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions," *Canadian Society for Mechanical Engineering International Congress*, 2023, pp. 1–6. (doi)

2. **A. Khabbazi**, R. Li, and J. Quinn, "Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid," *Canadian Society for Mechanical Engineering International Congress*, 2022, pp. 1–1. (doi)

1. **A. Khabbazi**, R. Li, and J. Quinn, "The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines," *International Green Energy Conference (IGEC-XIII)*, 2021, pp. 1–1. (doi)

Honors & awards

Selected

ASHRAE Graduate Student Grant-in-Aid Award. *(link)*. — ASHRAE, 2025

Best Paper Award at CSME 2023 Conference. *(link)*. — CSME, 2023

Best Presentation Award at CSME 2022 Conference, Advanced Energy Symposium. *(link)*. CSME, 2022

Others

Purdue University Graduate School "Say It In 6" Finalist. Purdue, 2024

UBC Graduate Scholarship. UBC, 2022

UBC Dean's Entrance Scholarship. UBC, 2021

1^{st} rank in GPA (4.0/4.0) among 155+ students. BS, 2016 – 2020

Conferences

8th International High Performance Buildings Conference, West Lafayette, USA 07/2024

ASHRAE Winter Conference, Chicago, USA 01/2024

Canadian Society for Mechanical Engineering 2023 Conference, Sherbrooke, Canada 05/2023

Canadian Society for Mechanical Engineering 2022 Conference, Edmonton, Canada 07/2022

13th International Green Energy Conference, Virtual 07/2021

Workshops & seminars

Intelligent Building Operations (IBO) Workshop, West Lafayette, USA 07/2024
What have we learned from field demonstrations of advanced commercial HVAC control? (Recording)

Industry experience

Summer Intern, Tesla (Reno, NV, United States) 05/2025 – Present

- Developed machine learning algorithms for real-time quality assessment on the Drive Unit team, reducing the need for destructive lab testing.
- Assisted with integrating a thermal imaging camera on the production line to enable real-time, non-destructive quality assessment.
- Attended weekly Inverter Engineering meetings to stay aligned with ongoing projects and deepen knowledge in advanced manufacturing and process optimization.

Student Researcher (Affiliate), FortisBC (Vancouver, BC, Canada) 09/2021 – 09/2023

- Supported FortisBC's technical assessments for major renewable gas projects, including input to the UBC H_2Lab construction.
- Worked directly with FortisBC's engineering team to translate research outcomes into practical recommendations for industry applications.

Research experience

PhD Research Assistant, Purdue University (IN, USA) 09/2023 – Present

- Assisted with commissioning Ray W. Herrick Laboratories as a testbed for advanced HVAC control by helping retrofit the Building Automation System (BAS) with Tridium Niagara and integrate ModBus for real-time monitoring. Contributed to developing occupant-facing dashboards for comfort and energy feedback.
- Built ML models to enable smart HVAC control at Herrick Labs, using CO_2 sensor data to detect occupancy with 98.3% accuracy. Automated data pipelines in Python and improved real-time analysis with InfluxDB and Linux, integrating results into dynamic energy management strategies.
- Designed and simulated HVAC control strategies using MPC and DeePC with MATLAB CVX and Python CVXPY, focusing on energy efficiency and occupant comfort. Compared MPC's model-based approach with DeePC's data-driven method, demonstrating that DeePC can achieve similar performance without a system model.
- Reviewed 100+ peer-reviewed publications on field demonstrations of MPC and RL for residential and commercial HVAC. Analyzed research trends using Python and visualization tools, leading to a published review paper.
- Modeled the thermodynamic performance of low-GWP refrigerants in heat pump systems for cold climates. Built cycle models in EES to analyze efficiency trade-offs and support sustainable HVAC design and decarbonization.

MS Research Assistant, University of British Columbia (BC, Canada) 07/2021 – 09/2023

- Led research on hydrogen injection into natural gas pipelines, including CFD analysis, real gas modeling in C, and CAD-based pipeline design.

- Managed and analyzed over 1 TB of simulation data, resulting in a journal paper and three conference presentations, receiving two conference awards.

Teaching experience

Graduate Teaching Assistant, University of British Columbia (BC, Canada) 09/2021 – 04/2023

- Co-taught core and advanced mechanical engineering courses, including Engineering Analysis I, Fluid Mechanics II Lab, and Heat Transfer Applications Lab.
- Earned an overall satisfaction rating above 80% in course evaluations.

Skills

Programming: Python, C, MATLAB, Markdown, Git, HTML, R

Machine Learning & Data Science: TensorFlow, Keras, NumPy, Pandas, scikit-learn, SciPy, Matplotlib, Seaborn

Data Management & Operating Systems: MySQL, Grafana, InfluxDB, Linux

Communication Protocols: Modbus, IO-Link, MTConnect

Engineering Tools: EES, ANSYS Workbench, OpenFOAM, Tecplot, SOLIDWORKS, CATIA, JMP

Selected courses

Thermal and Energy Systems: Distributed Energy Resources (ME597) — Analysis of Thermal Systems (ME518) — Advanced Thermodynamics (ME500) — Thermodynamics I&II — Refrigeration Systems — Power Plants — Heat Transfer I

Applied Mathematics & Data Science: Applied Machine Learning (ENGR418) — Statistical Methods (STAT511) — Advanced Mathematics I (MA527) — Numerical Computations

Automation, Control, and IoT: Introduction to Convex Optimization (AAE561) — Industrial IoT Implementation (ME597)

Fluids: Computational Fluid Dynamics (CFD) — Fundamentals of CFD — Multiphase Flows — Turbulence — Fluid Mechanics I&II

Academic services

Society memberships:

American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) 2023 – present
Canadian Society for Mechanical Engineering (CSME) 2021 – present

Reviewing:

8th International High Performance Buildings Conference, West Lafayette, USA 07/2024

Certifications

Unsupervised Learning, Recommenders, Reinforcement Learning Deep learning.AI

Advanced Learning Algorithms Deep learning.AI

Supervised Machine Learning: Regression and Classification Deep learning.AI

Version Control Meta

Introduction to Databases Meta

MySQL Fundamentals Tesla

AI Python Deep learning.AI

Applied Plotting, Charting & Data Representation in Python Coursera

Introduction to Data Science in Python Coursera

Python Data Structures Coursera